

CRYPTOGRAPHY COURSE

Scenario-based

CO2-AT-2

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1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.

B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

ANSWER:

Secure Data Transmission using Block Cipher

Problem Statement

A company is transmitting repeated financial data blocks over a network. Initially, they used **ECB (Electronic Code Book)** mode and noticed patterns in the ciphertext.

A. Explain Why This Issue Occurs

In **ECB mode**, each plaintext block is encrypted independently using the same encryption key.

If two plaintext blocks are identical, they produce the **same ciphertext block**. As a result, repeated patterns in the original data remain visible in the encrypted data.

Since financial records often contain repeated values, an attacker can identify these repeated patterns from the ciphertext. Although the attacker cannot directly read the data, the exposed patterns may reveal useful information, making ECB mode insecure for transmitting sensitive information.

B. Suggest a More Secure Block Cipher Mode and Justify Your Choice

The recommended block cipher mode is **CBC (Cipher Block Chaining) Mode**.

Justification

- CBC combines each plaintext block with the **previous ciphertext block** before encryption.
- It uses an **Initialization Vector (IV)** for the first block.
- Even if two plaintext blocks are identical, their ciphertext blocks will be different.
- This removes visible patterns in the ciphertext and provides better confidentiality.
- CBC is more secure than ECB for transmitting financial data because it protects sensitive information from pattern analysis.

Conclusion

- **Issue with ECB:** Identical plaintext blocks generate identical ciphertext blocks, exposing patterns.
 - **Recommended Mode: CBC (Cipher Block Chaining).**
 - **Reason:** CBC hides repeated patterns by using the previous ciphertext block and an Initialization Vector (IV), making it much more secure for transmitting confidential financial data.
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Final Answer

Question	Answer
Why does the issue occur?	ECB encrypts identical plaintext blocks into identical ciphertext blocks, revealing patterns.
Recommended Mode	CBC (Cipher Block Chaining)
Reason	CBC uses an Initialization Vector (IV) and the previous ciphertext block, producing different ciphertext even for identical plaintext blocks and providing better security.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- Identify the most suitable algorithm for this scenario.
- Justify your choice based on security strength and efficiency.

ANSWER:

Choosing an Encryption Algorithm for Banking System

Problem Statement

A banking application requires **high security** and **moderate performance** for encrypting customer transactions. The available encryption algorithms are **DES, 3-DES, and Blowfish**.

A. Identify the Most Suitable Algorithm

The most suitable algorithm for this banking application is **Blowfish**.

B. Justify Your Choice Based on Security Strength and Efficiency

DES

- Uses a **56-bit key**, which provides low security.
- Vulnerable to brute-force attacks.
- Not suitable for securing banking transactions.

3-DES

- Uses a **168-bit key**, providing stronger security than DES.
- Encrypts data three times, making it more secure.
- However, it is slower and requires more processing time.

Blowfish

- Uses a **128-bit key**, providing strong security.
 - Encrypts data faster than both DES and 3-DES.
 - Requires less processing time while maintaining high security.
 - Suitable for applications that require both security and efficient performance.
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Conclusion

Among the three algorithms, **Blowfish** is the best choice because it provides **strong security** with **high encryption speed**. It offers better performance than 3-DES while maintaining a high level of protection, making it suitable for banking applications where secure and efficient transaction processing is required.

Final Answer

Algorithm Security Performance Suitable for Banking

DES	Low	Moderate	No
3-DES	Very High	Slow	Good
Blowfish	High	Fast	Best Choice

Recommended Algorithm: Blowfish

Reason: Blowfish provides strong security, fast encryption, lower processing time, and efficient performance, making it the most suitable algorithm for encrypting customer transactions in a banking system.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

ANSWER:

Key Exchange in an Unsecured Network

Problem Statement

Two users want to establish a secure communication channel over a public network using the **Diffie-Hellman Key Exchange**. However, the network is vulnerable to attacks.

A. Explain How the Key Exchange Works

1. Both users agree on two public values:
 - A large prime number (**p**)
 - A generator (**g**)
2. User A selects a private key and computes a public key using:

Public Key = $g^{(\text{Private Key})} \bmod p$

3. User B also selects a private key and computes its public key using the same formula.
4. Both users exchange their public keys over the public network.

5. User A uses User B's public key and its own private key to compute the shared secret key.
 6. User B uses User A's public key and its own private key to compute the same shared secret key.
 7. Since both users obtain the same shared secret key, they can use it to encrypt and decrypt their communication securely.
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B. Identify a Possible Security Threat and Suggest a Solution

Security Threat

A **Man-in-the-Middle (MITM) Attack** is the main threat in Diffie-Hellman key exchange.

In this attack, an attacker intercepts the exchanged public keys and replaces them with fake public keys. As a result, both users unknowingly establish separate secret keys with the attacker instead of with each other. The attacker can then read, modify, and forward the messages without the users' knowledge.

Solution

The attack can be prevented by using:

- **Digital Signatures** to verify the identity of users.
- **Digital Certificates (PKI)** issued by a trusted Certificate Authority.
- **Authenticated Diffie-Hellman** or **TLS (Transport Layer Security)** for secure key exchange.
- Mutual authentication before establishing the shared secret key.

These methods ensure that the exchanged public keys are genuine and protect the communication from attackers.

Conclusion

Diffie-Hellman enables two users to establish a common secret key over a public network without transmitting the key itself. However, it is vulnerable to Man-in-the-Middle attacks if authentication is not used. Using digital signatures, certificates, or authenticated protocols such as TLS effectively prevents this attack.

Question	Answer
How does Diffie-Hellman work?	Both users generate private and public keys, exchange public keys, and independently compute the same shared secret key for secure communication.

Question	Answer
Security Threat	Man-in-the-Middle (MITM) Attack
Solution	Use Digital Signatures, Digital Certificates, Authenticated Diffie-Hellman, or TLS to authenticate the exchanged public keys and prevent attacks.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.

B. Explain how key distribution and management play a role in maintaining security.

ANSWER:

Public Key Cryptography for Email Security

Problem Statement

An organization plans to implement **RSA** for secure email communication.

A. Describe How RSA Can Be Used for Encryption and Decryption

RSA is a **public key cryptography** algorithm that uses two keys: a **public key** and a **private key**.

Encryption Process

1. Each user generates a pair of keys: a **public key** and a **private key**.
2. The sender obtains the receiver's **public key**.
3. The sender encrypts the email message using the receiver's public key.
4. The encrypted email is transmitted over the network.
5. Only the receiver can decrypt the email using their **private key**.

Decryption Process

1. The receiver receives the encrypted email.
2. The receiver uses their **private key** to decrypt the ciphertext.
3. The original email message is recovered and read securely.

This ensures that only the intended receiver can access the email contents, providing confidentiality.

B. Explain How Key Distribution and Management Play a Role in Maintaining Security

Proper key distribution and management are essential for the security of RSA.

- The **public key** should be distributed securely so that senders can encrypt messages correctly.
- The **private key** must remain secret and should never be shared with anyone.
- **Digital Certificates** issued by a trusted **Certificate Authority (CA)** help verify that a public key belongs to the correct user.
- Keys should be stored securely and replaced periodically to reduce security risks.
- If a private key is compromised, it should be revoked immediately and a new key pair should be generated.

Effective key management ensures that only authorized users can decrypt emails and protects the system from unauthorized access.

Conclusion

RSA provides secure email communication by encrypting messages with the receiver's public key and decrypting them with the receiver's private key. Proper key distribution, secure storage, digital certificates, and regular key management are essential to maintain confidentiality and prevent unauthorized access.

Final Answer

Question	Answer
How is RSA used for email security?	The sender encrypts the email using the receiver's public key, and the receiver decrypts it using their private key.
Role of Key Distribution and Management	Public keys must be distributed securely, private keys must remain confidential, and digital certificates and proper key management ensure secure communication.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication.

The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.

B. Recommend a suitable cryptographic approach and justify your answer.

ANSWER:

Cryptography for IoT Devices

Problem Statement

A smart home system uses **low-power IoT devices** that require secure communication. The designer must choose between **RSA** and **Elliptic Curve Cryptography (ECC)**.

A. Analyze the Constraints of IoT Devices

IoT devices have limited hardware resources and must operate efficiently. The main constraints are:

- **Limited Processing Power:** IoT devices have low-speed processors, so they cannot perform complex computations quickly.
 - **Limited Memory:** They have small RAM and storage capacity for storing keys and program data.
 - **Low Power Consumption:** Most IoT devices operate on batteries, so energy-efficient algorithms are required.
 - **Limited Bandwidth:** IoT devices often communicate over low-speed wireless networks, requiring smaller key sizes and less data transmission.
 - **Real-Time Communication:** Devices must process and exchange data quickly with minimal delay.
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B. Recommend a Suitable Cryptographic Approach and Justify Your Answer

The most suitable cryptographic approach is **Elliptic Curve Cryptography (ECC)**.

Justification

- ECC provides the **same level of security** as RSA while using **much smaller key sizes**.
- Smaller keys require **less memory** for storage.
- ECC performs encryption and decryption with **lower computational cost**.
- It consumes **less processor power and battery energy**, making it ideal for low-power devices.

- Faster cryptographic operations improve the performance of real-time IoT communication.
- ECC is widely used in smart home systems, wireless sensor networks, and other embedded devices.

Although RSA is secure, it requires larger key sizes and more processing power, making it less suitable for resource-constrained IoT devices.

Conclusion

For smart home IoT devices, **ECC is the preferred cryptographic algorithm** because it provides strong security with smaller keys, lower memory usage, reduced power consumption, and faster performance compared to RSA.

Final Answer

Question	Answer
Constraints of IoT Devices	Limited processing power, limited memory, low power consumption, limited bandwidth, and real-time communication requirements.
Recommended Cryptographic Approach	Elliptic Curve Cryptography (ECC)
Reason	ECC provides strong security with smaller keys, lower computational cost, reduced memory usage, lower power consumption, and better performance for IoT devices.